

PRANITA SANTOSH PANCHAL

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SUMMARY

Backend-focused Full-Stack Software Engineer with experience building secure REST APIs, distributed systems, and production-ready web applications using TypeScript, Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, Docker, and CI/CD. Skilled in designing scalable backend architectures, implementing authentication and encryption, and deploying containerized services. Passionate about backend engineering, DevOps, cloud infrastructure, and building reliable software for global remote teams.

TECHNICAL SKILLS

- Languages:** JavaScript (ES2022+), TypeScript, Rust, SQL, Python, Java, C++
- Backend:** Node.js, Express.js, FastAPI, REST API Design, JWT, OAuth 2.0, Microservices, Async I/O, Zod Validation
- Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- Databases:** PostgreSQL, MongoDB, Mongoose, Neo4j, MySQL
- DevOps & Cloud:** Docker, Docker Compose, Kafka, GitHub Actions, CI/CD Pipelines, Linux/Unix, AWS (basics), Vercel, Render
- Security & Tools:** AES-256 Encryption, Bcrypt, Role-Based Access Control (RBAC), Secure Secret Management, Git, GitHub, Postman, Prisma ORM

PROJECTS

SECUREVAULT BACKEND - Encrypted Secret Management Service | TypeScript, Node.js, Prisma, PostgreSQL, Docker, Terraform

- Built a SaaS secret management platform with secure REST APIs, JWT authentication, RBAC, AES-256 encryption, and PostgreSQL for multi-user credential storage.
- Designed a scalable backend architecture using Prisma and Docker, enabling secure namespace isolation, automated workflows, and production-ready deployment.

ARGUS-PRISM — Predictive Fraud Detection System | TypeScript, Node.js, Express.js, Kafka, Neo4j, PostgreSQL

- Real-time mule account detection pipeline that flags “warming” behavior 72 hours before illicit funds arrive, targeting ₹36,014 Cr in annual bank fraud.
- Architected 5 detection engines (graph analysis, behavioral scoring, taint tracking, STR generation, upstream threat mapping) over event-driven Kafka; cut regulatory evidence prep from 3–7 days to under 60 minutes.
- Built for iDEA 2.0 (Union Bank of India); selected as National-Level Finalist from 1,000+ teams across India.

JURISYNTH AI - High-Performance Legal Management Pipeline | Node.js, Express.js, MongoDB, React.js, JWT, OCR/Webservice APIs

- Architected a multi-tenant backend engine handling automated text parsing logic and multi-user data isolation across secure, token-authorized client sessions.
- Constructed asynchronous background task runner loops that cut web query processing bottlenecks, consistently delivering sub-200ms document and case version retrieval latencies.

EXPERIENCE

Web Developer Intern

Sept 2024 – Nov 2024

Cake4U (Startup)

Pune, Maharashtra

- Designed full-stack storefront workflows managing real-time catalogs, checkout baskets, and session state engines using Node.js and Express.js backends.
- Deployed asynchronous API endpoints for instant item updates, driving the business's first web order flow within two weeks of main branch release.

RESEARCH & ACHIEVEMENTS

iDEA 2.0 — National-Level Finalist | Union Bank of India

2026

- Selected as National-Level Finalist out of 1,500+ teams across India for ARGUS-PRISM, a pre-crime mule account detection system built on real 2026 RBI/FIU regulatory data.
- Presented to a panel of banking technology executives and regulators; recognized for innovation in financial fraud prevention.

Jurisynth: Engineering for Legal Clarity and Efficiency | IJSREM Journal (Volume 10, Issue 6) — Accepted 2026

2026

- Published primary co-author research introducing a domain-adaptive text processing framework combining system routing optimization, layout token normalization, and data ingestion pipelines; evaluated system constraints against real-world digital legal corpora to achieve a 0.34 compression ratio, 86% fact retention, and a 0.90 explainability score utilizing FAISS vector search

Education

PCET's Nutan College of Engineering and Research

2023 – 2026

B.Tech in Computer Science & Engineering - CGPA: 8.24

Pune, Maharashtra

Dr. D.Y. Patil Pratishthan's Y.B. Patil Polytechnic, Akurdi

2023

Diploma in Computer Science and Engineering — 84.46%

Pune, Maharashtra